

Minimal Grid Path

Time limit: 1.00 s

Memory limit: 512 MB

You are given an $n \times n$ grid whose each square contains a letter.

You should move from the upper-left square to the lower-right square. You can only move right or down.

What is the lexicographically minimal string you can construct?

Input

The first line has an integer n : the size of the grid.

After this, there are n lines that describe the grid. Each line has n letters between A and Z.

Output

Print the lexicographically minimal string.

Constraints

- $1 \leq n \leq 3000$

Example

Input	Output
4 AACA BABC ABDA AACA	AAABACA